

Datesight

Project Planning and Management

TEAM: 3

Table of Contents

Project Proposal 03

Project Plan 04

Task Assignment and Roles 05

Risk Assessment and Mitigation Plan 06

KPIS 08



Project overview

Project Name :	DATESIGHT
Date:	19 /10/2026
Prepared By:	TEAM 3
Supervisor:	MOATAZ BADRAN
Repository:	HTTPS://GITHUB.COM/MARYAMALBADAWY/DATESIGHT.GIT
Track:	PROFESSIONAL DATA ANALYSIS

Summary

DateSight is a data analysis and dashboard project that turns Egypt's date export data into clear, decision-ready views. The team collects and models trade data for 2016–2025, cleans and validates it, analyses growth, product mix, markets, and competitors, and delivers interactive Tableau dashboards as the final product.



problem statement

Egypt is the world’s largest producer of dates (about 1.86 million tonnes a year, FAO 2023, as cited by PEI Trade) yet exports only an estimated 35,000–40,000 tonnes, so its export share is far smaller than its production lead. Exporters, investors and policy makers do not have one consolidated, interactive view of how exports have grown, where they go, how Egypt compares with competitors, and where the opportunities are. The information is scattered across trade databases, news and reports, in different units and formats.

Objectives (SMART)

Why this project matters (External objectives): What decision-makers will get from DateSight.

E1. Show exporters and policymakers, in one simple dashboard, how Egypt's date exports grew from $41.4M in 2016 to $146.9M in 2025, and why they rose or fell. How we measure it: At least 8 clear insights, each backed by a figure and a source.

E2. Help them decide where to focus: which markets buy the most, and what to expect in 2026-2027. How we measure it: Top importing markets ranked, plus a simple forecast (low / base / high).

What we will deliver (Internal objectives): What the team builds to make that happen.

I1. Build a clean, reliable dataset. How we measure it: All data checks passed; every number has a known source. Due: 13 Nov I2.

I2. Build an easy-to-use Tableau dashboard (4-5 pages). How we measure it: Test users can find the top market and the growth trend without help. Due: 30 Nov I3.

I3. Submit all DEPI documents and code on GitHub, and present the project. How we measure it: 4 of 4 milestones on time. Due: 4 Dec

Scope

IN SCOPE

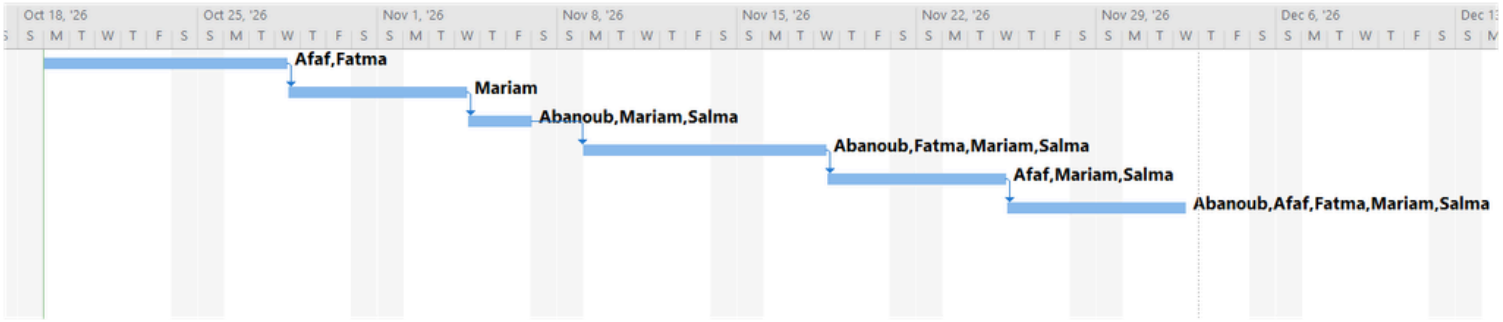
- Collection, modeling, cleaning and analysis of Egypt dates export data
- Competitor benchmarking
- Interactive Tableau dashboards
- Comparison of date varieties (only if reliable data is found)
- Optional/advanced: main obstacles to Egyptian dates exports (from cited studies)

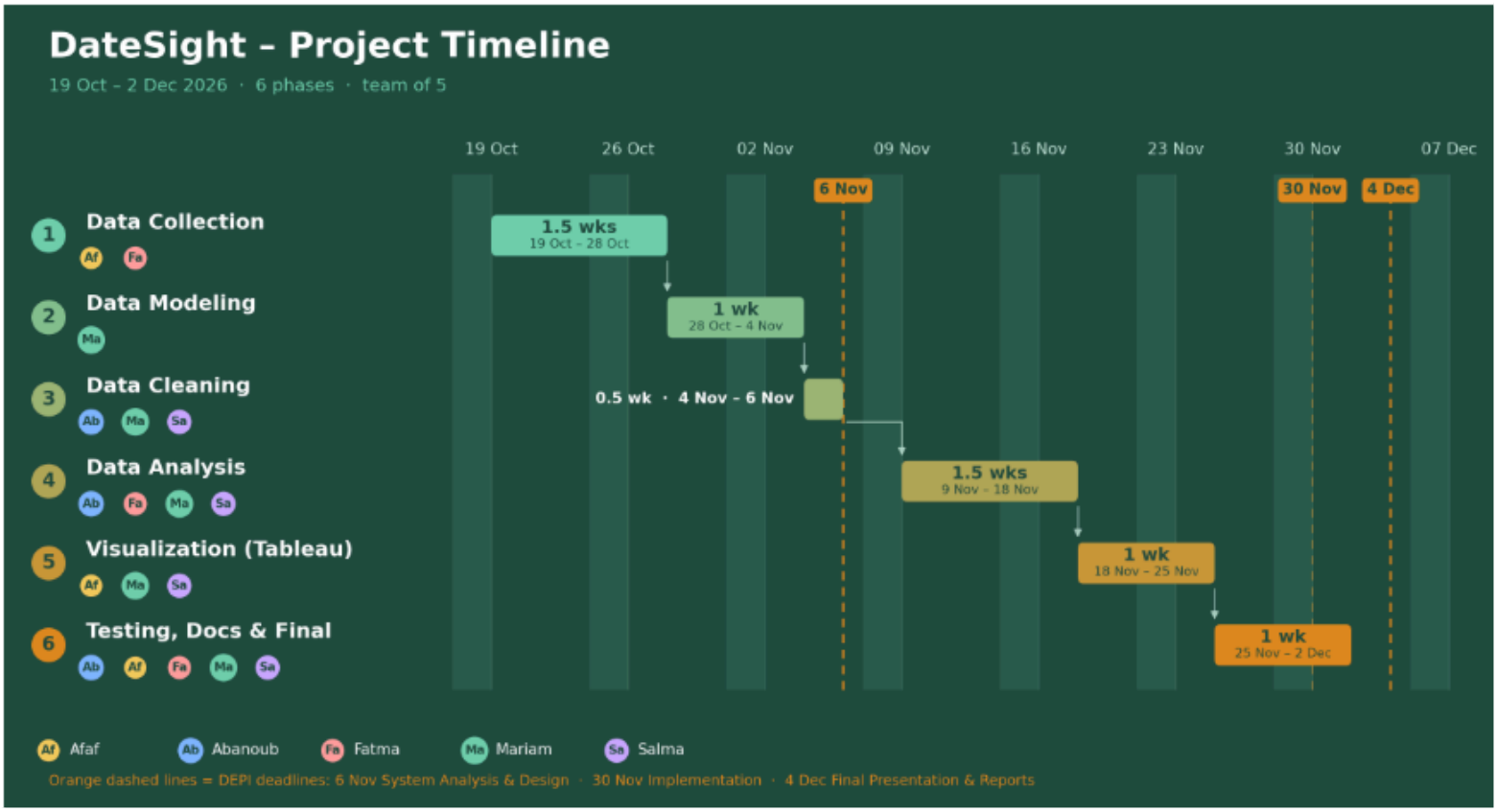
OUT OF SCOPE

- Web or mobile application, user accounts, payments
- Real-time data feeds
- Automated newsletter (listed as future work)
- Operational export advice for individual companies

Gantt chart

Task Name	Duration	Start	Finish	Predecessors	Resource Names
Data Collection	1.5 wks	Mon 10/19/26	Wed 10/28/26		Afaf,Fatma
Data Modeling	1 wk	Wed 10/28/26	Wed 11/4/26	1	Mariam
Data Cleaning	0.5 wks	Wed 11/4/26	Fri 11/6/26	2	Abanoub,Mariam,Salma
Data Analysis	1.5 wks	Mon 11/9/26	Wed 11/18/26	3	Abanoub,Fatma,Mariam,Salma
Visualization	1 wk	Wed 11/18/26	Wed 11/25/26	4	Afaf,Mariam,Salma
Testing, Docs & Final	1 wk	Wed 11/25/26	Wed 12/2/26	5	Abanoub,Afaf,Fatma,Mariam,Salma





Gantt chart


1
Data Collection
2
Data Modeling
3
Data Cleaning
4
Data Analysis
5
Visualization

Milestones

Date	DEPI Milestone	Team Work Leading to It	What We Submit
Oct 16, 2026	Project Planning & Management · Literature Review · Requirements Gathering	Planning, requirements & literature review (8 – 16 Oct)	Documents 1–3
Nov 6, 2026	System Analysis & Design	Data Collection (19 – 28 Oct) → Data Modeling (28 Oct – 4 Nov) → Data Cleaning (4 – 6 Nov)	Use cases, ERD, schema
Nov 30, 2026	Implementation (Execution)	Data Analysis (9 – 18 Nov) → Visualization (18 – 25 Nov)	Tableau dashboards, README, published link, GitHub history
Dec 4, 2026	Final Presentation, Testing & Reports	Testing, Docs & Final (25 Nov – 2 Dec)	Test plan/cases, user manual, technical documentation, presentation, infographic

Resource allocation

Resource	Allocation
People	5 team members (Afaf, Fatma, Salma, Mariam, Abanoub)
Tools	Python · Tableau · GitHub · document and diagram tools
Data Provided	Excel file + public trade and production databases (free tiers)
Time	About 7 weeks (13 OCT,4 DEC); weekly team meeting, weekly progress check, 1 review session before each DEPI milestone
Governance	GitHub repository with feature branches and pull-request reviews; one owner per deliverable

Task Assignment & Roles

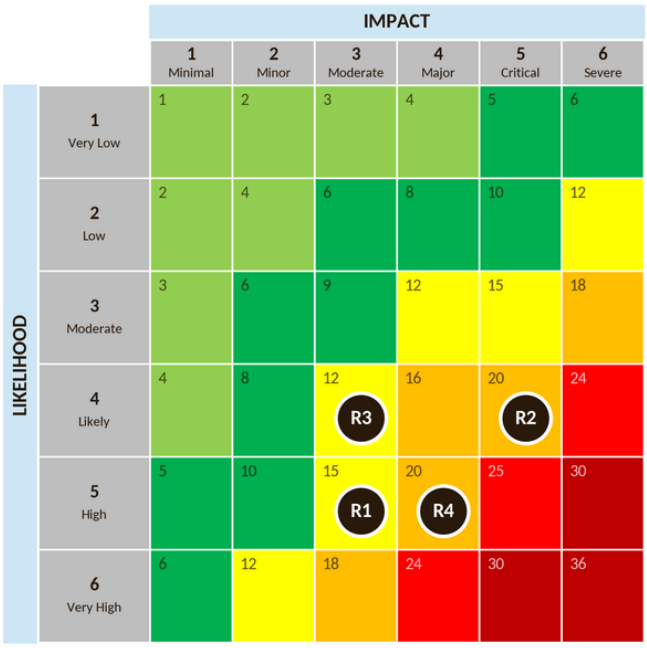
Member	Role	Main responsibilities
M1 Mariam	Project lead & data modeler	Planning and tracking, requirements, ERD and schema, data dictionary, document integration, GitHub owner, dashboard design lead
M2 Abanoub	Data cleaning	Transformations, validation checks, loading, data quality report, test cases, forecast back-tests
M3 Afaf	Data collection	Acquires trade and market data, keeps the Sources Register, prepares country/region geo-data for maps
M4 Salma	Data analyst & map/dashboard developer	Market and competitor analysis, concentration metrics, spatial views, Tableau dashboards
M5 Fatma	Domain expert & insights lead	Date varieties, seasonality and production context, literature review, validating insights, obstacles analysis, user acceptance testing

Risk Assessment & Mitigation

DEPI · PROFESSIONAL DATA ANALYSIS

DateSight — Risk Assessment

Risk score = Likelihood (1–6) × Impact (1–6)



ID	Risk & what it means	L	I	Score	Mitigation
R1	**Data shortage – date varieties** The export file reports dates under one code (HS 080410), so the varieties that were exported cannot be identified.	5	3	15 Moderate	Treat the varieties comparison as conditional; search export-council and trade-platform sources; if no reliable data, move it to future work and state the limitation.
R2	**Missing country-level data – importers & competing suppliers** The file has world totals only. Importing markets and competitor (supplier) exports must come from UN Comtrade / ITC Trade Map.	4	5	20 High	Download country-level and quantity data early; use the years available; if restricted, fall back to world-level analysis plus competitor benchmarking.
R3	**Dates market volatility** Exports swing sharply between years (−19% in 2017, +50% in 2018, +35% in 2022), which makes trends harder to read and explain.	4	3	12 Moderate	Show multi-year views and CAGR beside YoY; every explanation of a jump or drop needs a cited source, otherwise label it a hypothesis.
R4	**Forecasting difficulty – short, dated data** Only 10 annual points (2016–2025) are available, so forecasts for 2026–2027 are hard to make reliable.	5	4	20 High	Present scenarios as ranges, back-test on held-out years (MAPE ≤ 20%), and state the limitation clearly.

RISK LEVELS

Very Low (1–4)	Low (5–10)	Moderate (11–15)	High (16–20)	Very High (21–25)	Extreme (26–36)
No additional measures needed.	Acceptable; proceed with caution.	Acceptable if it cannot be reduced further; review.	Additional controls required before proceeding.	Critical; reduce the risk before proceeding.	Unacceptable; immediate action required.

Key Performance Indicators (KPIs)

Area	What we measure	Target	How we check
On-time delivery	Handing in every DEPI deliverable on its due date	4 out of 4	Compare submission dates with the plan
Data found	Getting export data by country and quantity for every year (2016–2025), or explaining clearly why some is missing	All years	Check the Data Sources Register
Data accuracy	Data passes all checks: totals match, no missing years, no negative numbers	100%	Validation report
Questions answered	Answering our 5 research questions, or explaining what data was missing	5 out of 5	Check the Insight log
Findings	Useful findings, each with a number, a source and what it means	At least 8	Check the Insight log
Forecast quality	Testing our forecast on past years, so we know how far off it can be	Off by 20% at most	Forecast notebook
Dashboard	Dashboard with 4–5 pages and at least 3 filters, and every number has a clear source	4–5 pages, 3+ filters, 100% sourced	Final Tableau workbook
Ease of use	At least 5 people try the dashboard and say it's easy to use and find the top market	80% satisfied, 90% find the answer	Short feedback form
Teamwork	Every member's work is visible on GitHub	5 out of 5	GitHub history

Literature Review ↗

Egyptian Dates Export Analytics & Dashboards

Prepared by :

DateSight Team

Prepared for :

DEPI – Professional Data Analysis

Table of Contents

01 Summary

02 Review Objectives

03 Scope and Method

04 Egypt’s Dates Sector in Brief

05 Related Studies

06 Existing Tools and Platforms

07 Methods Used in Similar Work

08 Gap Analysis

09 Suggested Improvements

10 Lecturer Feedback and Evaluation

11 References

01

Summary

“

This review looks at what has already been studied, built and measured about Egyptian dates exports, so DateSight starts from evidence instead of starting from zero. The literature agrees on the core problem: Egypt leads the world in production but holds a small share of exports. Most studies measure this with competitiveness indices published in papers, not with interactive dashboards. The sources also show weak data on date varieties and export figures that differ from one source to another. These findings shape how we collect, validate, analyse and forecast the data in the next phases.

15	6	6
sources reviewed	studies on Egyptian dates exports	gaps turned into project decisions

The 15 sources are 6 studies on Egyptian dates, 2 trade-data platforms, 3 industry and news sources, 1 sector overview and 3 papers on forecasting short series (full list in the References section).

02

Review Objectives

01

Understand what is already known about Egypt’s dates exports, markets and competitiveness.

02

Identify the data sources, tools and methods used in similar work.

03

Find the gaps that DateSight can fill.

04

Turn the findings into clear decisions for our data collection, modeling, analysis and dashboards.

> **How this review connects to the plan**
> Each finding in this document is linked to a phase of the DateSight plan: Data Collection, Data Modeling, Data Cleaning, Data Analysis, Visualization, and Testing, Docs and Final.

03

Scope and Method

→ Inclusion

- Studies and reports on Egyptian dates exports, competitiveness and markets.
- Trade-data platforms that cover dates (HS 0804).
- Industry and news sources on varieties, export figures and national strategy.
- Papers on forecasting short annual series.

→ Exclusion

- Studies about date palm farming techniques only, with no trade or market content.
- Company price lists and sales material.

→ Method

- We searched Egyptian agricultural-economics journals, trade platforms and industry news in October 2026.
- Each source was compared on three points: the data it uses, the method it applies and what it found.
- Limitation: several Arabic journal articles were reviewed through their published abstracts and summaries, so their details should be re-checked in the full text before final submission.

04

Egypt’s Dates Sector in Brief

Topic	What the sources say	Source
Production	Egypt is the largest date producer, with about 18% of world production in 2021 and about 1.65 million tonnes. [2] [9]	[2] [9]
Export share	Egypt’s share of world exports is small, but the figure changes by source: below 3% [2], about 3% [5], about 2.1% [3], below 4% in 2020 [11], and 4.5% in 2023 (8th exporter) [9].	[2] [3] [5] [9] [11]
Export value	About USD 43 million in 2020 [11]; USD 88.15 million in 2023 [9]; about USD 90 million a year, around 1% of agricultural exports [10].	[9] [10] [11]
Markets	63 importing countries in 2020 [11] and 71 markets in 2023 [9]. Morocco, Indonesia, Turkey and Russia are named as leading importers [10]; Morocco, Indonesia, Malaysia and Bangladesh are the markets studied in [1] and [2].	[1] [2] [9] [10] [11]
Varieties	In 2020 an official said only Siwi and Saidi were exported, and that Zaghloul and Samani were not exportable [11]. A later article says the current focus is premium types such as Medjool, Sukkary, Khalas and Ajwa, and that semi-dry types are only about 20% of production [10]. Medjool and Barhi are described as rare in Egypt [5]. An older sector overview groups Egyptian cultivars into fresh types such as Hayani, Zagloul and Samani, and half-dry types [15].	[5] [10] [11] [15]
Strategy	A national strategy for the date palm sector was set for 2024–2029 [9]. One article reports a target of USD 250 million export revenue within five years [10].	[9] [10]

Takeaway for DateSight

The same indicator, Egypt’s share of world dates exports, ranges from about 2% to 4.5% across sources. DateSight must record the source, year and definition of every figure and reconcile them with Trade Map data before using them in dashboards.

05

Related Studies

Study	Method	Main findings	What DateSight takes
[1] Arab Universities J. Agric. Sci. (2016)	Competitiveness indicators and econometric analysis of Egyptian dates exports.	Egypt has an advantage in market penetration in Morocco, Indonesia, Malaysia and Lebanon. The geographic concentration coefficient is close to zero, which the authors flag as a problem. Competitors had a price advantage.	Add market-share, concentration (HHI) and price comparison views.
[2] Assem et al. (2023)	Market share, market penetration rate and revealed comparative advantage (RCA), with scenarios to 2034 for four main markets.	Egypt has a significant comparative advantage. In a business-as-usual scenario its expected share of world imports drifts down, from about 2.4% in 2022 to about 1.6% in 2034.	Present forecasts as scenarios with ranges, not one number.
[3] Menoufia J. study (article 416789)	Assesses competitiveness and export efficiency, and the future of production and exports.	Egypt's exports are only about 2.1% of global exports despite its top production position.	Frame the analysis around the production-to-export gap.
[4] Egypt vs Tunisia (2021)	Balassa RCA index on fresh and dry dates markets.	Compares the two countries to propose ways of improving Egypt's marketing position in export markets.	Benchmark Egypt against a competitor; split by product type if data allow.
[5] New Valley case study	Competitiveness measures, geographic distribution of exports and export efficiency.	Egypt's share of world exports is about 3%. Medjool and Barhi are noted as rare varieties in Egypt.	Use as context for the varieties limitation.
[6] Abd El-Kawy et al. (2025)	Trends in the quantity, value and price of world dates exports, and Egypt's competitive position.	Places Egypt inside the world trend and lists problems facing date palm production.	Include a world-trend and price-per-tonne view.

Findings are taken from the abstracts and summaries of each study; see the Scope and Method section.

06

Existing Tools and Platforms

Tool	What it offers	Limits for our project	How we use it
ITC Trade Map [7]	Yearly trade flows for over 220 countries and territories and 5,300 products at 2, 4 and 6-digit HS level. Indicators include market share, market concentration and unit values. Time series since 2001, shown as tables, graphs and maps.	Some features need an account or a paid subscription. It covers all products, so it gives no date-specific analysis, and it does not separate varieties.	Main source for country-level exports and quantities, 2016–2025. Download early.
OEC [8]	Country and bilateral profiles with top products, trade value, annualized growth and economic complexity ranking.	General purpose. Egypt profiles group fruit into broad categories such as “Other Fruits”, so dates are not isolated.	Cross-check totals and use for context only.
Industry and news sources [9] [10] [11]	Factsheets and articles with export values, markets, varieties and strategy.	Secondary sources. Figures differ between sources and years (see section 04).	Use to explain events and context, and label as secondary.

Takeaway for DateSight
General trade platforms give the raw numbers but not a ready view for dates. DateSight adds a focused, interactive dashboard on HS 080410 built on validated data.

07

Methods Used in Similar Work

→ Measuring competitiveness

- Revealed comparative advantage (Balassa index) [4] and market penetration rate [1] [2].
- Market share of importing markets [1] [2] and geographic concentration [1].
- Price competitiveness against competing exporters [1] [5].
- These match the analysis planned for phase 4 of DateSight: YoY growth, CAGR, share of HS 0804, concentration (HHI), competitor benchmarking and price per tonne.

→ Forecasting short series

- Simple and Holt exponential smoothing performed well on small samples, especially for short-term forecasts [12].
- On short and unstable data with structural breaks, exponential smoothing was slightly worse at short horizons and better at longer ones; neural networks fit the past well but predicted poorly [13].
- On a longer quarterly series (1998–2017), ARIMA beat exponential smoothing and VAR, with the authors advising caution [14].

Takeaway for DateSight
With about 10 annual points (2016–2025), we prefer simple trend and smoothing models, avoid complex ones, back-test on held-out years and show scenario ranges. This supports our KPI of a back-test error of 20% or less.

08

Gap Analysis

Gap	Evidence	How DateSight responds
Static, general analysis	Reviewed studies publish indices in journal articles [1] [2] [3] [4] [5] [6]. Platforms give broad product data [7] [8].	Interactive Tableau dashboard (4–5 pages) with filters, KPIs and annotations, focused on HS 080410.
Thin data on varieties	An official named only two exported varieties in 2020 [11]; a later article names other premium types [10]. Our Excel file reports dates as one code.	Treat the varieties comparison as conditional; document the limitation (Risk R1).
Figures differ between sources	Export share ranges from about 2% to 4.5% (section 04).	Data Sources Register, validation checks, and a cited source for every dashboard figure.
Country-level data is hard to get	Trade Map has restricted and paid features [7].	Download early; fall back to world-level analysis plus competitors (Risk R2).
Sharp year-to-year swings need explanation	Our file shows −19% in 2017, +50% in 2018 and +35% in 2022, while the abstracts reviewed mostly report period-level indicators.	Insight log: every explanation needs a cited source, otherwise it is labeled a hypothesis (Risk R3).
Short series for forecasting	About 10 annual points [12] [13] [14].	Scenario ranges and back-tests on held-out years (Risk R4).

09

Suggested Improvements

01
Add the definition and year basis of each figure to the Data Sources Register.

02
Use Trade Map as the main country-level source and download it in the first week.

03
Add a price-per-tonne and competitor price comparison page, since price competitiveness appears in the studies.

04
Show forecasts as ranges with a back-test, not a single line.

05
Keep varieties as optional work unless reliable data is found.

→ Future work

- Variety-level analysis if reliable data appear.
- Add a revealed comparative advantage (RCA) view once country and quantity data are available.

10

Lecturer Feedback and Evaluation

To be completed by the lecturer.

Feedback and evaluation

Suggested improvements from the lecturer

11

References

[1] Comparative Advantage and Competitiveness of Egyptian Exports of Dates in the World Markets. *Arab Universities Journal of Agricultural Sciences*, Sep 2016. doi.org/10.21608/ajs.2016.14800

[2] Assem, A. et al. (2023). The Future Status of Egyptian Date Palm After the 2.5 Million Palm Trees National Project. Vol. 4, No. 1, pp. 20–27. ideas.repec.org/a/zib/zbsfna/v4y2023i1p20-27.html

[3] An Economic Study of the Future Expectations of the Production and Export of Egyptian Dates. *Menoufia Journal*, article 416789. mjabes.journals.ekb.eg

[4] Analysis of the Competitiveness of Egyptian and Tunisian Exports in the Fresh and Dry Dates Markets. *Scientific Journal of Agricultural Sciences*, 3(1), 2021, 227–236. sjas.journals.ekb.eg/article_173521.html

[5] Dates: A Case Study in the New Valley Governorate. jaess.journals.ekb.eg, article 199097.

[6] Abd El-Kawy, A. S., Shehata, G. A. B., El-Dalee, A. M. A. and Abd El-monem, A. (2025). The Economic Importance of Egyptian Foreign Agricultural Trade in Dates. *Journal of Agricultural Sciences and Sustainable Development*, 2(2), 192–203.

[7] International Trade Centre. Trade Map: Trade statistics for international business development. trademap.org

[8] Observatory of Economic Complexity (OEC). Egypt bilateral trade profiles. oec.world

[9] Arab Finance (21 Sep 2024). Egypt: A Global Leader in Date Production and Export. arabfinance.com/en/news/newdetails/18394

[10] FreshPlaza. Egypt aims to boost date exports amid global market growth. freshplaza.com/article/9714421

[11] News Central TV (23 Sep 2021). Egypt Exported Dates to 63 Countries in 2020 – Official. newscentraltv.com/?p=80144

[12] Andersson, J. (2006). Searching for the DGP when forecasting: is it always meaningful for small samples? *Economics Bulletin*, Vol. 3, EB-06C20070A.

[13] Comparative analysis of univariate time series modeling and forecasting techniques for short-term unstable data. essuir.sumdu.edu.ua/handle/123456789/68080

[14] Erkekoglu, H., Garang, A. P. M. and Deng, A. S. (2020). Comparative Evaluation of Forecast Accuracies for ARIMA, Exponential Smoothing and VAR. *International Journal of Economics and Financial Issues*, 10(6). doi.org/10.32479/ijefi.9020

[15] Riad, M. The Date Palm Sector in Egypt. *CIHEAM, Options Méditerranéennes*. ressources.ciheam.org/ressources/om/pdf/a28/96605879.pdf

DEPI · PROFESSIONAL DATA ANALYSIS

DateSight

Requirements Gathering — Stakeholders · User Stories & Use Cases · Functional & Non-functional Requirements

Item	Detail
Project	DateSight – Egyptian Dates Export Analytics & Dashboards
Programme / Track	DEPI – Digital Egypt Pioneers Initiative · Professional Data Analysis
Submission item	3. Requirements Gathering
Team	5 members: 1 BIS · 2 GIS · 1 Agriculture · 1 MIS
Version / Date	v1.0 · October 2026
Repository	GitHub (link to be added by the project lead)

1. Introduction

Purpose. This document captures what DateSight must do and how well, from the point of view of the people who will use or depend on it. It feeds the System Analysis & Design milestone (6 Nov 2026).

Product. Interactive Tableau dashboards on Egypt's dates exports (HS 080410), 2016–2025, supported by a documented, validated data model.

How requirements were gathered. Analysis of the provided export dataset and of public trade reports and platforms; review of similar export dashboards; team workshops across the five backgrounds (BIS, GIS, Agriculture, MIS); DEPI project guidelines. Needs of external stakeholders are *derived from desk research* and will be validated with real users during user acceptance testing.

> **Priority scale (MoSCoW):** Must = required for acceptance · Should = important, included if time allows · Could = nice to have. **Conditional** = depends on obtaining the necessary data (country-level exports, quantities, varieties).

2. Stakeholder Analysis

Stakeholder	Type	Needs and expectations	Influence	Interest	How we engage
Egyptian dates exporters	Primary user	Trend and growth of exports, best markets, competitors, price benchmarks, opportunities	Medium	High	Design the dashboard around their questions; test with exporters or export-minded users
Ministry of Agriculture and Land Reclamation	Primary user / policy	National export performance, market diversification, gap between production and exports	High	High	Executive overview and recommendations; share final report
Export councils and trade-support bodies (e.g., Agricultural Export Council, Food Export Council)	Primary user / influencer	Sector statistics for members and reports, ability to filter and export views	High	High	Use their published figures as cross-checks; offer exportable views
Date farmers, producers and cooperatives	Secondary user	Demand signals, price trends, which varieties and markets are growing	Low	High	Plain-language insights; variety view if data allows

Stakeholder	Type	Needs and expectations	Influence	Interest	How we engage
Investors, packers and processors	Secondary user	Growth rates, market concentration and risk, opportunity assessment	Medium	Medium	CAGR, HHI and scenarios clearly explained
Data providers (UN Comtrade, ITC Trade Map, FAOSTAT; national statistics and customs bodies)	Data supplier	Correct citation and respect of terms of use	Medium	Low	Cite every source; record licences in the Sources Register
Importers and foreign buyers	Indirect	Reliable view of Egyptian supply and trends	Low	Medium	Consider English labels and clear definitions
Researchers, students and analysts	Secondary user	Transparent method, reusable data definitions	Low	Medium	Methodology page, data dictionary, GitHub repository
DEPI instructors and evaluation committee	Evaluator	Complete documents, working dashboards, clear contribution of each member, GitHub evidence	High	High	Deliver every milestone; keep documentation and commit history clear
Project team (BIS, MIS, 2 × GIS, Agriculture)	Builder	Clear scope, realistic plan, reliable data	High	High	Weekly meetings, shared task board, pull-request reviews

Influence / interest summary

Manage closely (high influence, high interest)
Ministry of Agriculture · export councils · DEPI evaluators · project team

Keep informed (lower influence, high interest)
Dates exporters · farmers and cooperatives · investors and processors

Keep satisfied (medium influence, lower interest)
Data providers

Monitor (lower influence and interest)
Importers · researchers and students

3. User Personas (illustrative)

Persona	Goal	Key questions
Export manager at a dates company	Decide which markets and products to focus on	Where is demand growing? Who am I competing with? What price level is realistic?
Policy analyst (ministry or export council)	Monitor national performance and dependency risk	How fast are exports growing? Are we too reliant on a few markets? What may 2026–2027 look like?
Investor / agribusiness analyst	Assess attractiveness of dates exports	What is the growth rate and volatility? What are the risks?

4. User Stories

ID	As a...	I want to...	So that...	Acceptance criteria	Priority
US-01	Exporter	see how total dates exports changed from 2016 to 2025	I understand the momentum of the sector	Line chart + YoY %; values match the source data	Must
US-02	Policy analyst	see KPIs for the latest year (value, YoY, CAGR, share of HS 0804)	I can read the situation in seconds	Four KPI cards on the overview page; definitions in tooltips	Must
US-03	Any user	filter by year range and product	I can focus on what matters to me	Filters update all views on the page within 3 s	Must

ID	As a...	I want to...	So that...	Acceptance criteria	Priority
US-04	Policy analyst	compare dates with other HS 0804 products	I can see the role of dates in the category	Stacked/100% chart and share table	Must
US-05	Researcher	read the data sources, definitions and limitations	I can trust and reuse the analysis	Methodology page lists sources, units, dates, limitations	Must
US-06	Exporter	see the top importing markets and their share Conditional	I can choose where to focus	Ranked top-10 list or map with shares; year filter	Should
US-07	Policy analyst	see market concentration (HHI) Conditional	I can judge dependency on a few markets	HHI value and explanation for the selected year	Should
US-08	Exporter	compare Egypt with competing exporters Conditional	I know where Egypt stands	Value, share and growth for Egypt vs main competitors	Should
US-09	Exporter	see average price per tonne over time Conditional	I can benchmark my price	Price = value ÷ quantity; trend line; units stated	Should
US-10	Policy analyst	view 2026–2027 scenarios	I can plan support measures	Low / base / high scenarios with assumptions and error range	Should
US-11	Export council analyst	export a view as image or PDF	I can use it in my own reports	Export option available; layout stays readable	Should
US-12	Any user	read a short list of key insights and recommendations	I get the message in 30 seconds	≥ 8 insights with figure, source and implication	Should
US-13	Farmer or exporter	compare date varieties Conditional	I know which varieties sell best	Variety comparison with a cited data source	Could
US-14	Researcher	see the main obstacles to Egyptian dates exports	I understand what limits growth	Short summary with cited studies (optional/advanced)	Could
US-15	DEPI evaluator	open the GitHub repository with documents, code and dashboard link	I can assess completeness and each member's work	README, folder structure, commit history, deployed link	Must

5. Use Cases

5.1 Use case list

ID	Use case	Primary actor	Related stories	Priority
UC-01	View executive overview	Exporter, policy analyst, investor	US-01, 02, 12	Must
UC-02	Filter and explore trends and product mix	Any user	US-03, 04	Must
UC-03	Analyse markets and concentration Conditional	Exporter, policy analyst	US-06, 07	Should
UC-04	Benchmark against competitors Conditional	Exporter, investor	US-08, 09	Should
UC-05	Compare date varieties Conditional	Exporter, farmer	US-13	Could
UC-06	View forecast scenarios	Policy analyst, investor	US-10	Should
UC-07	Read methodology and data sources	Researcher, any user	US-05	Must
UC-08	Export or share a view	Export council analyst	US-11	Should
UC-09	Refresh and validate data	Data analyst (team / maintainer)	Supports all	Must

5.2 Key use case descriptions

Field	UC-01 – View executive overview
Actor	Exporter, policy analyst or investor
Preconditions	Dashboard published; data validated
Main flow	1. User opens the dashboard. 2. System shows KPI cards (latest-year value, YoY, CAGR, share of HS 0804) and the 2016–2025 trend. 3. User hovers a point to read the value and definition. 4. User reads the key-insight panel.
Alternative / exception	If the data fails validation, the page shows a data-status warning instead of KPIs.
Postcondition	User understands the headline performance of dates exports.

Field	UC-03 – Analyse markets and concentration
Actor	Exporter or policy analyst
Preconditions	Country-level export data loaded and validated

Field	UC-03 – Analyse markets and concentration
Main flow	1. User opens the Markets page. 2. System shows the top importing countries (map and ranked bars) with share. 3. User selects a year. 4. System updates shares and the HHI value.
Alternative / exception	If country-level data is missing for the selected year, the page states the limitation.
Postcondition	User identifies main markets and dependency risk.

Field	UC-06 – View forecast scenarios
Actor	Policy analyst or investor
Preconditions	Historical series validated; forecast method documented
Main flow	1. User opens the Forecast page. 2. System shows history and low/base/high scenarios for 2026–2027. 3. User reads assumptions and error range.
Alternative / exception	If the forecast fails its back-test threshold, the page labels it “indicative only”.
Postcondition	User understands plausible export paths and their uncertainty.

Field	UC-09 – Refresh and validate data
Actor	Data analyst
Preconditions	New or corrected source data available
Main flow	1. Analyst places the raw file in data/raw. 2. Pipeline transforms the data (units, HS level, country names). 3. Validation checks run (totals reconcile, no missing years, no negatives). 4. Analyst reloads the Tableau extract and republishes.
Alternative / exception	If a check fails, the load stops and the issue is logged.
Postcondition	Dashboard shows validated data with an updated source note.

6. Functional Requirements

ID	Name	Requirement	Priority	Source	Note
FR-01	Data import	The solution ingests the cleaned export datasets (2016–2025) into the data model.	Must	US-01	
FR-02	Data validation	Data passes validation checks (HS-level totals reconcile, no missing years, no negative values) before use in dashboards.	Must	UC-09	
FR-03	KPI cards	Overview page shows latest-year export value, YoY growth, CAGR and share of HS 0804.	Must	US-02	
FR-04	Trend view	Line chart of annual export value 2016–2025 with YoY change.	Must	US-01	
FR-05	Filters	Filters for year range and product; country filter where data exists.	Must	US-03	
FR-06	Product mix	Composition of HS 0804 by product, with the share of dates and a ranked table.	Must	US-04	
FR-07	Growth diagnostics	YoY changes with annotations for key events, each backed by a cited source.	Should	US-01	
FR-08	Market analysis	Top importing countries (map and ranked bars) with share and HHI.	Should	US-06, 07	Conditional
FR-09	Competitor benchmarking	Egypt vs main exporters: value, world share and growth.	Should	US-08	Conditional
FR-10	Price analysis	Average export price per tonne over time, with units stated.	Should	US-09	Conditional
FR-11	Variety comparison	Compare date varieties by export or sales contribution.	Could	US-13	Conditional
FR-12	Forecast scenarios	Low / base / high scenarios for 2026–2027 with assumptions and error range.	Should	US-10	
FR-13	Insights panel	At least 8 documented insights and recommendations, each with figure, source and implication.	Should	US-12	
FR-14	Interactivity	Tooltips with definitions, highlight and drill-down actions.	Should	US-03	
FR-15	Methodology page	Lists data sources, units, definitions, update dates and limitations.	Must	US-05	
FR-16	Export / share	Users can export a view as image or PDF.	Should	US-11	
FR-17	Navigation	Consistent navigation between all dashboard pages.	Must	US-01–04	
FR-18	Obstacles summary	Short summary of the main obstacles to Egyptian dates exports, from cited studies.	Could	US-14	Optional
FR-19	Repository & docs	Code, data (where permitted), documents and the dashboard link are in GitHub with a README.	Must	US-15	

7. Non-functional Requirements

ID	Category	Requirement	How verified
NFR-01	Performance	Each dashboard page renders in ≤ 5 s and filters respond in ≤ 3 s on a normal broadband connection.	Timed tests
NFR-02	Accuracy & integrity	All displayed totals reconcile to the source; 100% of validation checks pass before publishing.	Validation report
NFR-03	Traceability	Every metric has a documented definition and source; no figure appears without one.	Data dictionary audit
NFR-04	Usability	A first-time user finds the top market and the growth trend in ≤ 2 minutes without training; UAT satisfaction ≥ 80%.	UAT with ≥ 5 users
NFR-05	Accessibility	Text contrast ≥ 4.5:1; meaning is never conveyed by colour alone; readable font sizes.	Design checklist
NFR-06	Compatibility	Works in current Chrome, Edge, Firefox and Safari, on desktop and tablet; mobile layout reviewed.	Cross-browser test
NFR-07	Reliability & availability	Dashboards reachable through a published link; data extracts and workbooks versioned so any version can be restored.	Link check, GitHub history
NFR-08	Security & privacy	Only public, aggregated data; no personal data; data licences and terms of use respected.	Sources Register review
NFR-09	Maintainability	Modular pipeline with a data dictionary and README; adding a new year takes ≤ 1 day.	Documented refresh test
NFR-10	Scalability	The star schema accepts new years, products and countries without redesign.	Schema review
NFR-11	Language	Dashboard labels and documents in English (project language); Arabic labels are a future option.	Design review

8. Assumptions, Constraints and Dependencies

Assumptions

- Public trade and production data can be downloaded and cited
- Tableau (Desktop or Public) is available to the team
- The team meets weekly

Constraints

- DEPI deadlines: 16 Oct, 6 Nov, 30 Nov, 4 Dec 2026
- Annual data, 10 years only (2016–2025)
- The provided file has world totals only and no varieties

> **Known data gap.** The export dataset does not identify date varieties, and has no importing countries or quantities. Requirements FR-08 to FR-11 are therefore **conditional**: they are built only if reliable data is found; otherwise the limitation is documented and the item moves to future work.

9. Traceability: Research Questions → Requirements

Research question	Requirements	Dashboard page
RQ1 Growth and its drivers	FR-03, 04, 07, 13	Overview · Growth diagnostics
RQ2 Share of dates in HS 0804	FR-06	Product mix
RQ3 Markets and competitors	FR-08, 09	Markets & competitors
RQ4 Price per tonne	FR-10	Markets & competitors
RQ5 Scenarios 2026–2027	FR-12	Forecast & recommendations
Extra: varieties · obstacles	FR-11 · FR-18	Product mix · Insights

10. Open Points

- Identify a reliable source for date varieties and prices (owner: Agriculture member).
- Confirm availability of country-level and quantity data from UN Comtrade / ITC Trade Map (owner: GIS member).
- Validate stakeholder needs with at least five real or representative users before the 30 Nov implementation milestone.